Jordan Watkins, PhD

Curriculum Vitae

(708) 256-5771 · incohatus.none@gmail.com · github.com/incnone · LinkedIn · Charlottesville, VA · relocating

Research interests: differential geometry; dynamics on Riemannian manifolds; rank rigidity and manifolds without focal points. Secondary interests in Bayesian modeling and optimization, developed through applied work.

EDUCATION

Ph.D. in Mathematics · University of Michigan — 2013

Specialization: differential geometry (dynamics on Riemannian manifolds). Thesis: "The Rank Rigidity Theorem for Manifolds With No Focal Points."

B.A. in Mathematics and Physics · Knox College — 2007

Valedictorian

PUBLICATIONS

"The Higher Rank Rigidity Theorem for Manifolds With No Focal Points" · *Geometriae Dedicata*, vol. 164 — Aug 2012 · pp. 319–349

Springer · arXiv:1111.5428

RESEARCH

Doctoral thesis · University of Michigan — 2013

"The Rank Rigidity Theorem for Manifolds With No Focal Points" — an extended treatment of the published higher-rank rigidity result.

TEACHING EXPERIENCE

Graduate Student Instructor, Mathematics · University of Michigan — 2007–2011

- Independently taught multiple sections of Calculus I and Calculus II — lecturing, assessment, and grading
- Teaching assistant for Differential Equations
- Led required weekly tutoring sessions in single- and multivariable calculus, differential equations, and basic probability
- Mentored individual paying students, including in linear algebra

Mathematics & Physics Tutor · Independent — Intermittently since 2005

- Tutored across a wide range: topology, abstract algebra, differential equations, statistics, linear algebra, and electricity & magnetism
- Supported computer-science students in C++
- Graduate-school teaching included weekly tutoring in calculus, differential equations, and probability, plus independent paying students

SOFTWARE & QUANTITATIVE PROJECTS

Competitive Rating & Matchmaking Systems · CoNDOR · open source

multiple seasons, through 2022

github.com/incnone/condor_ratings

A Bayesian rating model, series-outcome forecasting, and automated matchmaking, rebuilt and refined across several league seasons (Python, C++, MySQL, PuLP)

- Built and iterated on a Bayesian paired-comparison rating model: per-player Gaussian priors combined with binomial match evidence, fit by MAP estimation
- Derived the log-likelihood gradient analytically and wrote the optimizer by hand — batch gradient ascent with L2 convergence tests on both parameter change and gradient norm, iteration and wall-clock limits, and log-likelihood tracing
- Selected the prior variance by maximizing log-likelihood on held-out matches from a previous season — model selection against real data rather than hand-tuning
- Judged a normal model's tails too thin for real performance variation and designed a heavier-tailed link function — a mixture of an error function and a rational sigmoid whose tail weight is a hyperparameter fit to past match data ($\alpha = 0.42$) — calibrated to agree with the Bradley–Terry model at a chosen reference rating
- Patched Rémi Coulom's C++ Bayesian Elo engine (bayeselo) to expose the covariance matrix it computes internally, then modeled two players' ratings as jointly normal so series predictions account for correlated rating uncertainty
- Converted ratings into calibrated win probabilities and full best-of-3 / best-of-5 outcome distributions, exported as JSON for downstream use
- Automated weekly matchmaking as an integer linear program (PuLP/CBC) maximizing an information-theoretic objective — the entropy of the predicted result, penalized by rating gap — under per-player degree constraints
- Added a local-search pass performing minimum-cost pairwise exchanges to enforce a minimum cycle length in the matchup graph, with unit tests

Co-Developer — *Bean and Nothingness* · RBOR Games

2020–2021 · shipped Dec 2021

store.steampowered.com/app/1706090/Bean_and_Nothingness/

Logic-puzzle game · 98% positive from 198 Steam reviews · prototype work from 2012

- Co-lead programmer (C++, SDL, OpenGL): replay/undo state system, GUI level editor shipped via Steam Workshop, custom sound mixer, core game-loop and entity logic
- Lead puzzle designer: created 100+ and edited all ~200 puzzles

Architect & Maintainer — Necrobot · Open source

2015–Present

github.com/incnone/necrobot

Discord bots managing competitive races and league seasons; in continuous community use for 10+ years

- One layered codebase (~150 modules, 16 packages, documented acyclic dependencies) running two production bots
- Async Python with MySQL persistence, Google Sheets integration, and TrueSkill-based (Bayesian) ratings

CONTINUED EDUCATION

Mathematical Physics Seminar · University of Virginia

2024–Present

Regular participant. Ongoing engagement with current research in geometry and mathematical physics.

Erdős Institute — Data Science Boot Camp · coursework

Applied data science in Python (pandas, scikit-learn).

Self-directed study in machine learning (PyTorch), AI ethics, and post-doctoral mathematics and physics; ~4,000 reputation and 86 answers on Mathematics Stack Exchange.

SERVICE & COMMUNITY

- Long-term maintainer and community moderator for the open-source Necrobot project and its competitive community (10+ years)
- Policy and moderation on large public Discord servers

SKILLS

Mathematics	Differential geometry, linear algebra, dynamics, probability, Bayesian inference
Programming	C++, Python, SQL
Tools	Git, MySQL, PuLP, SDL/OpenGL, Google Sheets API

Career note: independent study and freelance tutoring 2013–2020, during part of which I carried family caregiving responsibilities.